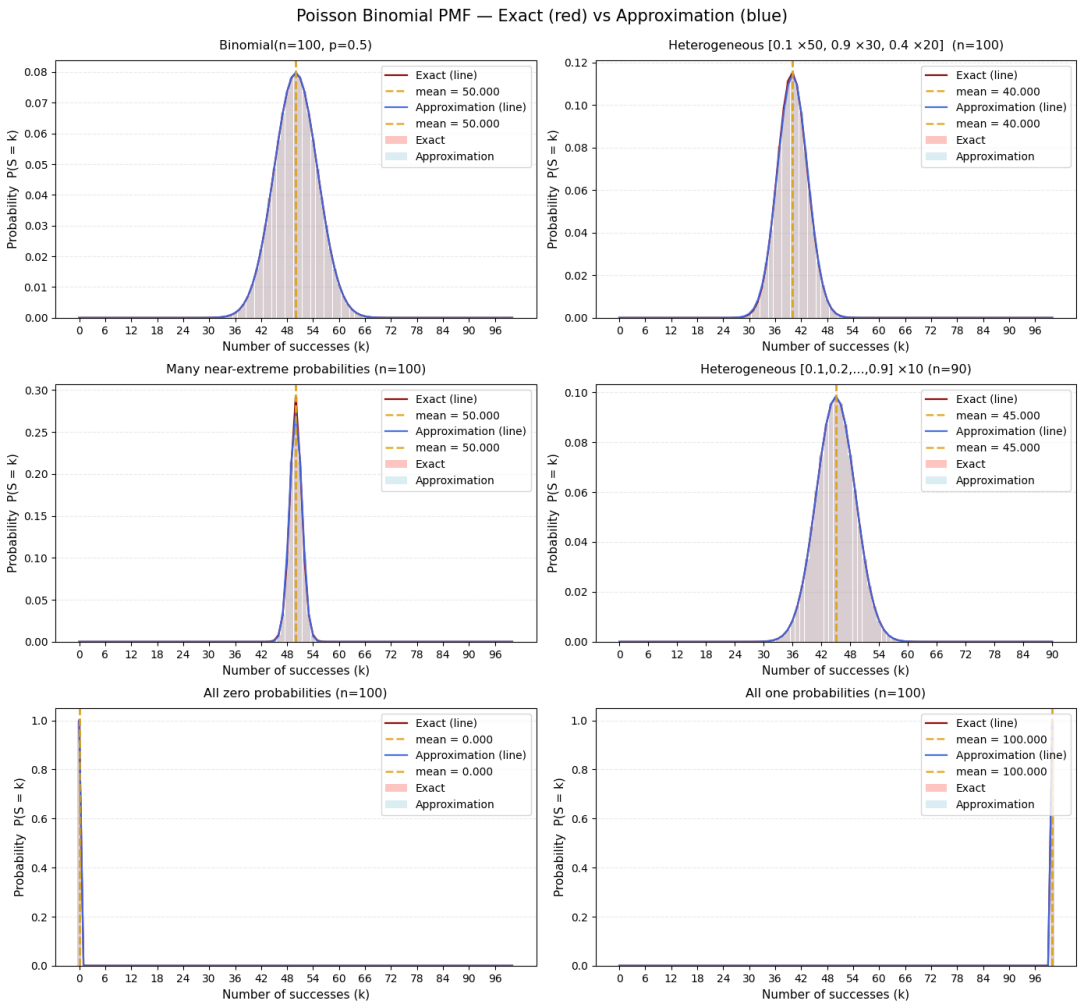
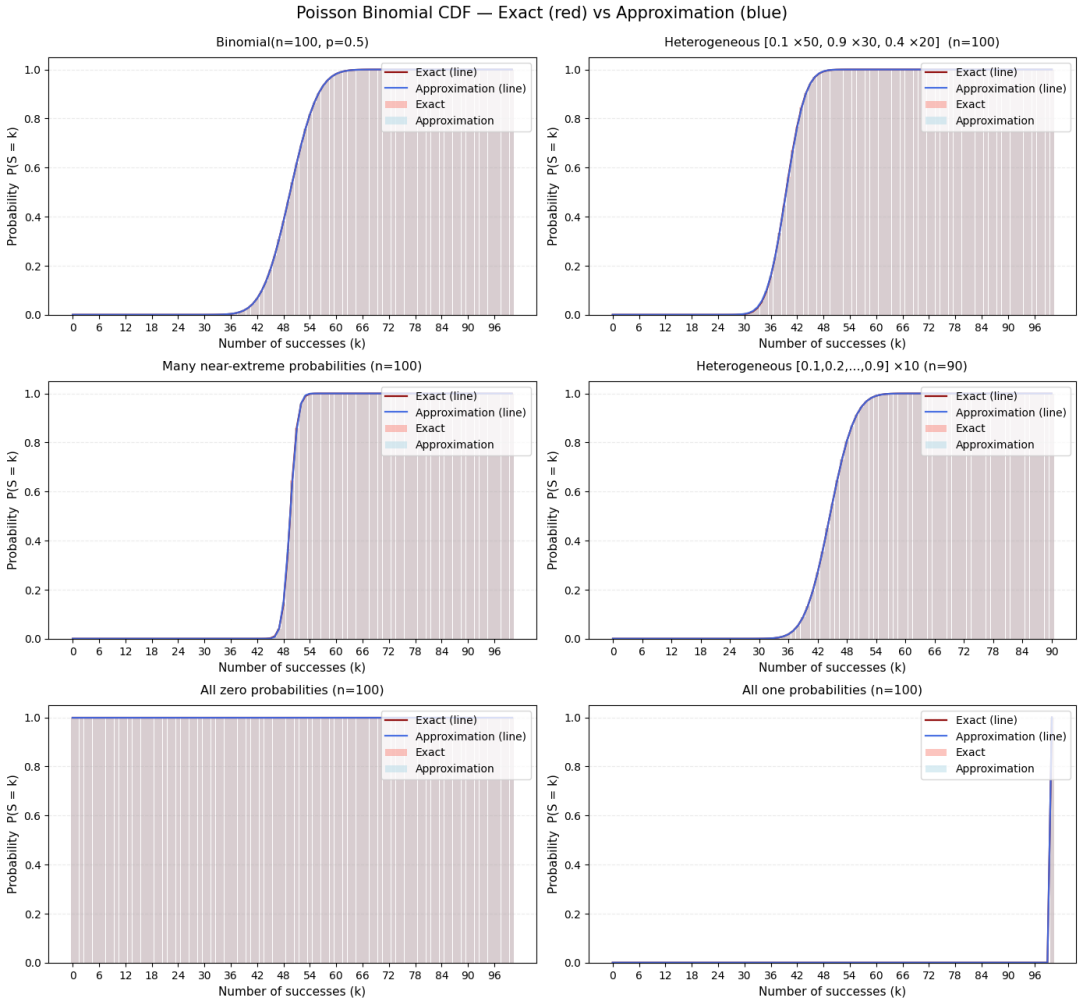
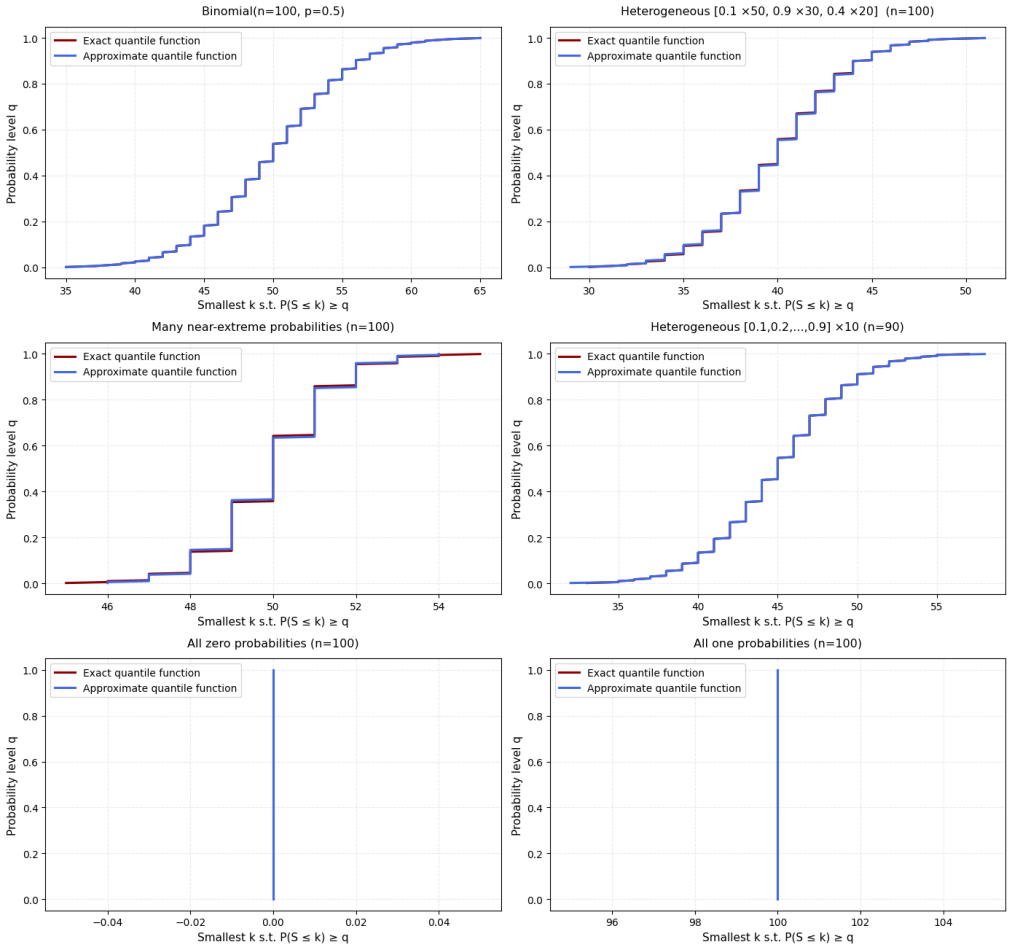

Poisson Binomial PMF — Exact (red) vs Approximation (blue)
Binomial(n=100, p=0.5)
Heterogeneous [0.1 ×50, 0.9 ×30, 0.4 ×20] (n=100)
Many near-extreme probabilities (n=100)
Heterogeneous [0.1,0.2,...,0.9] ×10 (n=90)
All zero probabilities (n=100)
All one probabilities (n=100)
Exact (line)
mean = 50.000
mean = 40.000
mean = 45.000
mean = 0.000
mean = 100.000
Approximation (line)
Exact
Approximation
Probability P(S = k)
Number of successes (k)


Poisson Binomial CDF — Exact (red) vs Approximation (blue)
Binomial(n=100, p=0.5)
Heterogeneous [0.1 ×50, 0.9 ×30, 0.4 ×20] (n=100)
Many near-extreme probabilities (n=100)
Heterogeneous [0.1,0.2,...,0.9] ×10 (n=90)
All zero probabilities (n=100)
All one probabilities (n=100)
Exact (line)
Approximation (line)
Exact
Approximation
Probability P(S = k)
Number of successes (k)

Poisson Binomial Quantile Function (ICDF) — Exact vs Approximation